Optimizing Formula 1 Pit Stop Strategy using Deep Learning

A project report submitted in partial fulfillment of the requirements for the degree

of

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By

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Abstract

Formula 1 pit stop strategy is one of the most consequential real-time decisions in motorsport. Teams rely on proprietary simulation software to recommend optimal pit lap timing, a capability unavailable to independent analysts. This project investigates whether a deep learning model trained on publicly available lap time data can recommend pit stop windows with competitive accuracy. A Gated Recurrent Unit (GRU) architecture is trained on 2022–2023 race data to predict lap-to-lap tire degradation, and a simulation framework compares the projected cost of staying out versus pitting for each candidate lap. This project conducts five ablation experiments and three multi-phase strategy refinements evaluating them against 122 race-driver combinations from the 2024 Formula 1 season. The final system achieves 52.5% strategy accuracy (within ± 5 laps of actual pit timing), a +21.5% improvement over a multi-stop brute-force baseline. Compound-specific modeling—training separate GRUs for SOFT, MEDIUM, and HARD tire compounds—is the single largest contributor, accounting for +26.2% of accuracy gain. Per-driver accuracy ranges from 40% to 70%, revealing that the absence of driver and team identity features is the primary remaining bottleneck.

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1. Introduction

1.1 Problem

Formula 1 race strategy is determined in real time by a team of engineers who must decide exactly when a driver should pit for fresh tires. This decision is among the highest leverage in motorsport: a well-timed pit stop can gain 5–10 seconds of net track position through an undercut, while a poorly timed stop can cost a podium. The optimal lap to pit depends on a multitude of factors such as tire degradation rate, remaining fuel load, ambient temperature, track position relative to competitors, and expected race pace on different compounds.

Professional teams invest millions of dollars in proprietary simulation software that models these interactions with telemetry data, real-time competitor tracking, and historical tire behavior. This capability is entirely unavailable to independent researchers, broadcasters, and fans. No publicly available tool suggests reliable, quantitative pit stop recommendations grounded in data.

1.2 Project Statement

This project develops and evaluates a deep learning system that recommends Formula 1 pit stop windows by learning tire degradation patterns from historical lap time data and simulating the net time cost of pitting at each candidate lap.

1.3 Approach

The approach consists of three primary components: (1) a data pipeline built on the FastF1 API [1] that extracts lap time, tire compound, fuel load, and weather features from 2022–2023 race data; (2) a GRU sequence model [3] trained to predict the next lap's tire degradation given the past 10 laps of race context; and (3) a simulation framework that uses the trained model to project race time under two futures, to decide whether the driver should stay on worn tires or pit for fresh tires. This decision is made at each candidate lap, identifying the first lap where pitting becomes net positive.

The full experimental arc spans nine model configurations, two data pipeline variants, and five strategy-level phases. Both positive and negative results are documented to characterize which design choices matter most.

1.4 Organization of this Report

Section 2 covers the relevant background: F1 tire degradation, sequence modeling with recurrent networks, and related work. Section 3 describes the system architecture including the data pipeline, model designs, and strategy simulator. Section 4 presents the methodology, ablation results, and strategy evaluation across all phases. Section 5 summarizes contributions and future work.

2. Background

2.1 Key Concepts

2.1.1 Formula 1 Tire Degradation and Pit Stop Strategy

Formula 1 regulations require each driver to pit atleast once, and use at least two different tire compounds during a dry race. The three primary compounds (SOFT, MEDIUM, and HARD) differ in grip level and durability. SOFT tires offer peak grip and are considered the fastest, however, only for $\sim$15–25 laps before significant degradation; HARD tires are considered the slowest but last $\sim$35–50 laps and generate less grip throughout. Furthermore, tire degradation is influenced by numerous factors such as track temperature, compound type, driving style, etc. As a stint progresses, lap times increase monotonically due to tire wear, providing a time-series signal that, in theory, should predict when pitting becomes beneficial.

However, the pit stop decision is not just about when tires degrade. It involves strategic considerations such as the undercut (pitting early to gain position through fresher tires), the overcut (staying out while a competitor pits, maintaining track position), and the safety car window (pitting under a safety car to take a "free" stop). This project addresses the tire degradation component

only; competitor position and safety car timing are treated as out-of-scope.

2.1.2 Recurrent Neural Networks for Time Series

Long Short-Term Memory (LSTM) networks [2] and Gated Recurrent Units (GRUs) [3] are the dominant architectures for sequential prediction tasks. Both maintain a hidden state that summarizes prior context, updated at each timestep through gating mechanisms.

LSTM cells use three gates (input, forget, output) and a separate cell state, giving them the concept of 'memory', and strong capacity for long-range dependencies at the cost of higher parameter counts. GRU cells use two gates (reset, update) and no separate cell state, making them faster to train and less prone to overfitting on moderate-sized datasets. For short sequences (10–20 laps), the memory advantage of LSTM is less relevant, and GRU has been shown to match or outperform LSTM in similar settings[4].

Attention mechanisms[5] extend recurrent models by computing a weighted sum over all hidden states rather than using only the final state, allowing the model to selectively focus on the most informative timesteps. Their benefit is most pronounced on sequences of 20 or more steps where individual timesteps carry distinct signals.

2.1.3 Delta Prediction for Degradation Modeling

Standard sequence models trained to predict absolute lap time face a systematic bias: the model must first "explain" the circuit's baseline lap time before it can model the degradation delta on top of it. This makes it difficult to learn compound-specific degradation slopes.

Reformulating the target as $\Delta t = t_{n+1} - t_n$ (lap-to-lap change) decouples the degradation signal from the baseline, allowing the model to focus exclusively on the rate of change. This formulation is directly motivated by degradation physics, which suggests that tire wear increases lap time at a rate that depends on compound hardness and track temperature, not on the absolute lap time.

2.2 Related Work

2.2.1 Race Strategy Simulation

Commercial F1 strategy tools (e.g., McLaren Electronic Systems, Williams Racing Analytics) are widely reported to use physics-based tire models combined with Monte Carlo simulations over opponent strategies. These systems require real-time telemetry and proprietary tire compound data, neither of which are available for public use.

Academic work on F1 strategy is severely limited due to the lack of publicly available data. Heilmeier et al. [7] develop a race simulation framework for circuit motorsports that incorporates tire degradation, fuel load, and safety car probability, using a parametric tire model fitted on historical timing data. Their approach requires compound-specific parameters typically sourced from team data and focuses on pre-race strategy planning rather than real-time window detection.

Tulabandhula and Rudin [8] apply machine learning to sports decision problems more broadly, demonstrating that data-driven approaches can approximate expert judgment in complex sequential settings. Reinforcement learning has also been informally explored for racing strategy, but these approaches require simulator environments that are difficult to calibrate without access to official team data.

2.2.2 Tire Degradation Modeling

Tire degradation has been modeled as a parametric function of stint length in physics-based simulators [7]. Data-driven approaches in the broader motorsport literature include regression over lap time residuals and recurrent models trained on historical sector times. To my knowledge, no published work applies compound-specific GRU models to pit stop window detection using publicly available lap data.

2.2.3 Publicly Available F1 Data

The FastF1 Python library [1] provides a clean interface to the official Formula 1 Livetiming API, including lap times, tire compound, weather, and driver information for all races since 2018. It is the only fully open data source for F1 lap timing and is used in this project as the sole data source.

3. Architecture

3.1 High-Level Design

The system consists of three main components arranged in a sequential pipeline:

1. **Data pipeline:** Fetches historical lap data via FastF1, applies compound and track status filters, engineers features, and normalizes per race.
2. **GRU model (per compound):** Three independent GRU networks trained on sequences filtered by tire compound (SOFT, MEDIUM, HARD).
3. **Strategy simulator:** At inference time, loads a 2024 race, generates worn and fresh tire predictions for each candidate pit lap, and identifies the first lap where pitting is net positive.

3.2 Data Pipeline

3.2.1 Data Source and Filtering

Race data is fetched using FastF1 v3.8.2 for the 2022 (22 rounds), 2023 (22 rounds), and 2024 (24 rounds) seasons. The 2024 season is held out entirely for evaluation; models are trained exclusively on 2022–2023 data. The model is trained on a temporal split between seasons rather than a random split across all seasons in order to prevent overfitting. Races within the same season share car performance and trajectories and regulatory context, so random sampling would allow the model to learn season specific patterns that would inflate the accuracy results without reflecting true generalization. For example, if the model is trained on car data from some races in 2024, then

it could use the learned pattern from 2024 to perfectly predict the pit stop timing for the rest of the 2024 races that reside in the evaluation set.

Three filters are applied to each race:

1. `IsAccurate == True`: removes laps with timing errors
2. `TrackStatus == '1'`: retains only green-flag laps (removes safety car, yellow flag, and red flag laps)
3. `PitInTime.isna()`: removes the pit-entry lap where the driver lifts off

Only dry-compound laps (SOFT, MEDIUM, HARD) are retained, and wet condition tires such as 'intermediates' and 'wet' are excluded. Weather data (air temperature, track temperature) is merged onto each lap via `pd.merge_asof` on lap start time.

3.2.2 Feature Engineering

The input feature vector for each lap consists of 9 features:

Table 1: Input Features

Feature	Description	Normalization
LapTimeSeconds	Lap time (s)	Per-race MinMax
StintLength	Laps on current tire set	Per-race MinMax
FuelLoad	Estimated fuel remaining (kg)	Per-race MinMax
AirTemp	Ambient air temperature (°C)	Per-race MinMax
TrackTemp	Track surface temperature (°C)	Per-race MinMax
Compound_SOFT	1 if on SOFT tires	Binary
Compound_MEDIUM	1 if on MEDIUM tires	Binary
Compound_HARD	1 if on HARD tires	Binary
TrackTemp_global	TrackTemp scaled across all training races	Global MinMax

Fuel load is estimated as $\max(0, 110 - \text{LapNumber} \times 1.6)$ kg, using a standard burn rate of 1.6 kg/lap from a 110 kg fuel load.

Continuous features are then normalized to $[0, 1]$ on a per-race basis using scikit-learn's `MinMaxScaler`:

$$\tilde{x}_i^{(r)} = \frac{x_i^{(r)} - x_{\min}^{(r)}}{x_{\max}^{(r)} - x_{\min}^{(r)}} \tag{1}$$

where r indexes the race and $x_{\min}^{(r)}, x_{\max}^{(r)}$ are computed over all laps in that race. Per-race scaling accounts for the wide variation in baseline lap times across circuits. Without it, the model would have to first learn circuit identity before learning degradation. The trade-off is that absolute temperature differences across circuits are erased. The `TrackTemp_global` feature is therefore added separately, and then normalized once across all training races rather than per race, to preserve that signal.

3.2.3 Sequence Building

For each driver stint, a sliding window of width 10 generates training samples. A window starting at lap i produces input $X_i \in \mathbb{R}^{10\times 9}$ and target $y_i = t_{i+10} - t_{i+9}$ (the lap-to-lap delta of the next lap). Stints shorter than 10 laps produce no samples and are discarded. After splitting by compound, the training dataset contains:

Table 2: Training Sequences by Compound (2022–2023)

Compound	Train Sequences	Test Sequences (2024)
SOFT	4,087	200
MEDIUM	9,853	442
HARD	15,041	2,584
Total	28,981	3,226

3.3 Model Architecture

The project evaluates two primary recurrent architectures: TireLSTM and TireGRU. Both take a sequence of shape $(B, 10, 9)$ as input and produce a scalar delta prediction.

TireGRU (selected final architecture). At each timestep t, the GRU computes a reset gate $\mathbf{r}_t$,

an update gate $\mathbf{z}_t$, a candidate hidden state $\tilde{\mathbf{h}}_t$, and the new hidden state $\mathbf{h}_t \in \mathbb{R}^{64}$:

$$\mathbf{r}_t = \sigma(W_r \mathbf{x}_t + U_r \mathbf{h}_{t-1} + \mathbf{b}_r) \tag{2}$$

$$\mathbf{z}_t = \sigma(W_z \mathbf{x}_t + U_z \mathbf{h}_{t-1} + \mathbf{b}_z) \tag{3}$$

$$\tilde{\mathbf{h}}_t = \tanh(W_h \mathbf{x}_t + U_h(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}_h) \tag{4}$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t \tag{5}$$

where σ is the logistic sigmoid and $\odot$ denotes element-wise multiplication. After 10 timesteps, the final hidden state is projected to a scalar prediction:

$$\hat{y} = \text{Linear}(\text{Dropout}(\mathbf{h}_{10})) \in \mathbb{R} \tag{6}$$

Configuration: hidden size 64, 2 stacked layers, dropout 0.2, batch-first. Total parameters: 39,425.

The model is trained to minimize mean squared error on the lap-to-lap delta target:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2 \tag{7}$$

Optimization uses the Adam optimizer [6] with learning rate 10^{-3}, ReduceLROnPlateau scheduler (patience=3, factor=0.5), gradient clipping at norm 1.0, and early stopping at patience=7 on validation loss.

An LSTM with a learned attention pooling layer was also evaluated. A linear projection scores each of the 10 LSTM hidden states, a softmax over the timestep axis produces attention weights, and the context vector is the weighted sum of the hidden states before the output projection:

$$\alpha_t = \text{softmax}_t(\mathbf{w}^\top \mathbf{h}_t) \tag{8}$$

$$\mathbf{c} = \sum_{t=1}^{10} \alpha_t \, \mathbf{h}_t, \quad \hat{y} = \text{Linear}(\text{Dropout}(\mathbf{c})) \tag{9}$$

This is a content-based attention pooling rather than the additive query–key formulation of [5]:

there is no separate query vector and no $\tanh$ nonlinearity. This variant was the worst performer (MAE -22.5% vs. baseline), as discussed in Section 4.3.

3.4 Strategy Simulator

At inference time, the simulator loads a 2024 race for a specific driver, normalizes the data using the same per-race scaler, and evaluates each candidate pit lap as follows:

Worn tire projection (`predict_worn`): Starting from the last 10 observed laps, the model auto-regressively predicts 20 future laps. Because the GRU outputs a delta $\Delta\hat{t}_i$ rather than an absolute lap time, predicted lap times are reconstructed by integration:

$$\hat{t}_{i+1} = \hat{t}_i + \Delta\hat{t}_i \tag{10}$$

seeded from $\hat{t}_0$ equal to the last observed lap time. At each rollout step, StintLength increments by 1 and FuelLoad decrements by the per-lap burn rate; the compound one-hot remains constant.

Fresh tire projection (`predict_fresh`): Same integration scheme, but seeded from the earliest low-tire-life laps of the target compound (normalized StintLength < 0.2, i.e. within the first 20% of the longest stint observed in the race) earlier in the race, with StintLength reset to 0 to simulate a fresh stint.

Decision criterion: For each candidate pit lap ℓ, compute:

$$\Delta(\ell) = \underbrace{L_{\text{pit}} + \sum_{k=1}^{20} \hat{t}_k^{\text{fresh}}}_{\text{pit cost}} - \underbrace{\sum_{k=1}^{20} \hat{t}_k^{\text{worn}}}_{\text{stay cost}} \tag{11}$$

where L_{pit} is the normalized pit-stop time loss (per-circuit pit-loss seconds divided by the maximum lap time). A 22-second default is used when no per-circuit value is supplied, but the 2024 calendar uses circuit-specific values that range from roughly 16 s (Las Vegas) to 28 s (Singapore), with around half the rounds deviating by at least 2 s from the default—notably Singapore, Imola, and Monaco at the high end and Las Vegas, Canada, Azerbaijan, and Saudi Arabia at the low end.

Since L_{pit} defines the threshold a stop must overcome to be net-positive, a uniform value would systematically over-recommend stops at slow pit-lane circuits and under-recommend them at fast ones, shifting the optimal pit window by several laps. The recommended pit lap is the first ℓ where $\Delta(\ell) < 0$. One-stop and two-stop strategies are evaluated by brute-force search over all valid (p_1) and (p_1, p_2) pit lap pairs; the total race time under each strategy determines which is preferred.

4. Methodology, Results, and Analysis

4.1 Methodology

4.1.1 Evaluation Metric

The primary evaluation metric we used is strategy accuracy. The fraction of race-driver combinations where the system's predicted first pit lap is within ± 5 laps of the nearest actual pit lap. Formally, for the set of evaluated race-driver pairs R, with predicted first pit $\hat{p}_r$ and nearest actual pit p_r^*:

$$\text{StratAcc} = \frac{1}{|R|} \sum_{r \in R} \mathbb{1}[\, |\hat{p}_r - p_r^*| \leq 5 \,] \tag{12}$$

The ± 5 lap tolerance reflects the real variance in F1 team strategy decisions since the same race scenario is often handled 5–10 laps differently by different teams, making strict lap accuracy an unreasonably tight criterion [7].

MAE on the held-out 2024 lap time sequences is reported as a secondary metric:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |\hat{y}_i - y_i| \tag{13}$$

but is shown across multiple experiments to be a poor proxy for strategy accuracy (see Section 4.3).

Two no-parameter baselines are used. The *moving average baseline* (used for absolute lap-time targets) predicts each lap's time as the mean of the 10 preceding laps:

$$\hat{y}_i^{\text{base}} = \frac{1}{10} \sum_{k=1}^{10} t_{i-k}. \tag{14}$$

The *mean-delta baseline* (used for delta targets, including the compound-specific results in Table 5) predicts the next lap-to-lap change as the mean of the 9 lap-to-lap deltas observed within the input window:

$$\widehat{\Delta y_i}^{\text{base}} = \frac{1}{9}\sum_{k=1}^{9}(t_{i-k} - t_{i-k-1}). \tag{15}$$

All MAE improvements are reported relative to whichever baseline matches the model's training target.

4.1.2 Test Set

The 2024 Formula 1 season (24 rounds) is held out entirely for strategy evaluation. Models see no 2024 data during training or hyperparameter tuning. The final expanded evaluation covers 6 drivers—VER (Verstappen, Red Bull), NOR (Norris, McLaren), LEC (Leclerc, Ferrari), SAI (Sainz, Ferrari), HAM (Hamilton, Mercedes), RUS (Russell, Mercedes)—across all 24 rounds, producing up to 144 race-driver combinations. Combinations where a driver retired or did not make a pit stop under green-flag conditions are automatically skipped, yielding 122 valid evaluations.

4.2 Results

4.2.1 Architecture Comparison

Table 3 summarizes all model configurations evaluated across three data versions. The GRU architecture (hidden=64, layers=2, dropout=0.2) is the consistent winner across all data configurations, and is selected as the base for all subsequent strategy-level experiments.

Figure 1 shows training and validation loss curves for the best LSTM and GRU configurations from Run 1. The GRU converges in 21 epochs versus 30+ for LSTM, and its train to validation gap is visibly narrower, consistent with lower overfitting risk at this dataset size. Figure 2 shows the test MAE for the three main Run 1 configurations: the GRU achieves a 40% reduction over the moving average baseline while the default LSTM slightly exceeds baseline MAE.

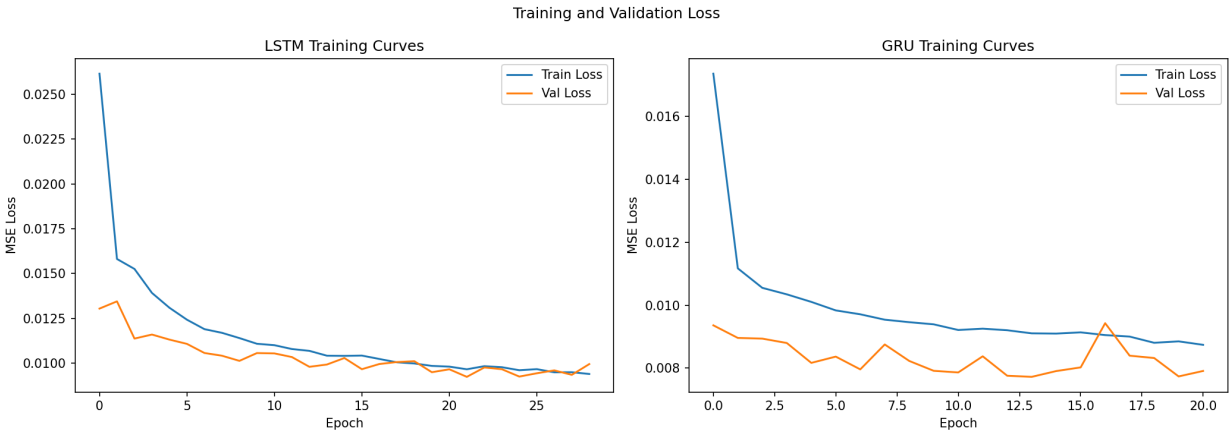


Figure 1: Training and validation loss curves for the best LSTM (left) and GRU (right) configurations (Run 1). The GRU converges faster and shows a tighter train–validation gap, indicating less overfitting.

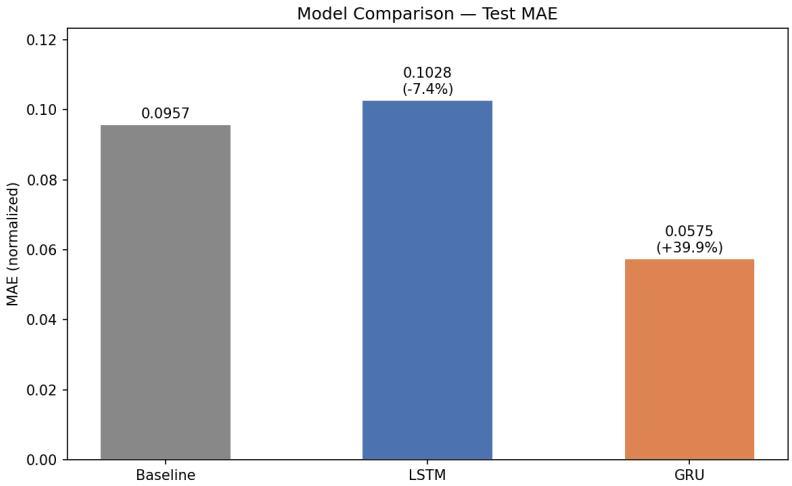


Figure 2: Test MAE for the baseline, best LSTM, and GRU on Run 1 (2023 data, 14k sequences). The GRU achieves a 39.9% improvement over the moving average baseline; the default LSTM slightly underperforms it.

Table 3: MAE Results Across Architectures and Data Versions

Architecture	Config	MAE	vs. Baseline	Stopped Epoch
Run 1 — 2023 only (∼14k sequences), baseline MAE = 0.0957				
LSTM	hidden=64, dropout=0.2	0.0914	+4.5%	34
LSTM	hidden=128, dropout=0.2	0.1158	−20.9%	32
LSTM	hidden=64, layers=3, dropout=0.3	0.1016	−6.2%	30
LSTM	hidden=64, dropout=0.4	0.0656	+31.5%	31
GRU	**hidden=64, dropout=0.2**	**0.0575**	**+39.9%**	**21**
Run 2 — 2022+2023 (28k sequences), baseline MAE = 0.0957				
LSTM	hidden=64, dropout=0.2	0.1044	−9.0%	32
LSTM	hidden=128, dropout=0.2	0.1092	−14.1%	33
LSTM	hidden=64, layers=3, dropout=0.3	0.0819	+14.5%	31
LSTM	hidden=64, dropout=0.4	0.0698	+27.1%	33
LSTM + Attention	hidden=64, dropout=0.2	0.1173	−22.5%	28
GRU	**hidden=64, dropout=0.2**	**0.0589**	**+38.5%**	**24**
Run 3 — 2022+2023, green-flag filter only, baseline MAE = 0.0605				
LSTM	hidden=64, dropout=0.2	0.1116	−84.4%	—
LSTM	hidden=64, dropout=0.4	0.0464	+23.3%	—
GRU	**hidden=64, dropout=0.2**	**0.0363**	**+40.0%**	**—**

4.2.2 Strategy Accuracy Progression

Table 4 traces strategy accuracy across all five phases of strategy-level development. The largest single jump is Phase 2 (+26.2 percentage points), driven by splitting training by tire compound. Phase 3 and Phase 4 both regressed and were reverted; the final system uses the Phase 2 compound-specific configuration evaluated over 122 race-driver combinations.

Table 4: Strategy Accuracy Across All Phases

Phase	Key Change	Evaluations	Accuracy
Baseline (single-stop, shared GRU)	—	42	31.0%
Phase 1: Multi-stop simulation	Brute-force 2-stop search	42	35.7%
Phase 2: Compound-specific GRUs	Separate model per compound	42	61.9%
Phase 3: Min first-pit constraint	Hard minimum lap threshold	42	59.5% (reverted)
Phase 4: Longer sequence (seq=15)	Window length 10→15 laps	42	50.0% (reverted)
Phase 5: 6-driver eval	**Expanded to 122 evaluations**	**122**	**52.5%**

4.2.3 *Compound-Specific Model Results (Phase 2)*

Separate GRU models are trained on sequences filtered by the compound active at each timestep. Table 5 shows training results for the three compound-specific models. The HARD model benefits most from the compound split (14.1% lower MAE than the mean-delta baseline), likely because hard-compound stints are longer and the degradation signal is more gradual, giving a homogeneous training distribution the most leverage.

Table 5: Phase 2 Compound-Specific GRU Training Results

Model	Train Sequences	Test MAE	vs. Mean-Delta Baseline
gru_delta_soft	4,087	0.042113	−10.2%
gru_delta_medium	9,853	0.037233	−10.9%
gru_delta_hard	15,041	0.035449	−14.1%

4.2.4 *Per-Driver Accuracy*

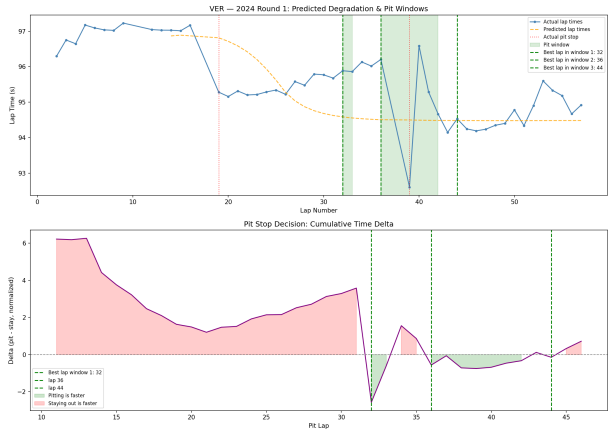
Table 6 shows strategy accuracy broken down by driver in the full 6-driver evaluation. The 30% spread across drivers is the single most important finding for understanding system limitations.

Table 6: Per-Driver Strategy Accuracy (Phase 5, 122 evaluations)

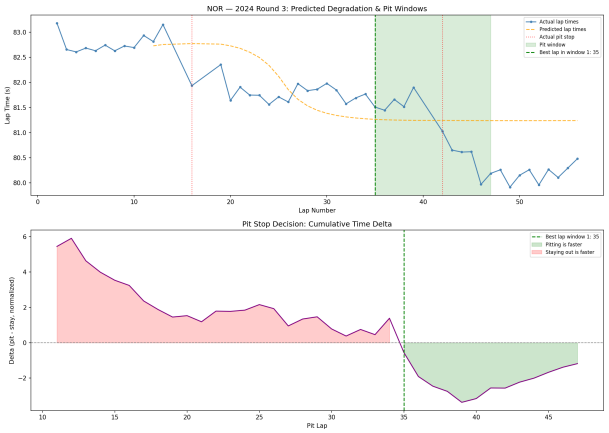
Driver	Team	Correct	Evaluated	Accuracy	Median Error (laps)
RUS	Mercedes	14	20	**70.0%**	4.0
SAI	Ferrari	12	20	60.0%	4.5
VER	Red Bull	11	21	52.4%	5.0
NOR	McLaren	10	21	47.6%	11.0
LEC	Ferrari	9	20	45.0%	7.0
HAM	Mercedes	8	20	40.0%	11.5
Overall		**64**	**122**	**52.5%**	**6.5**

Figure 3 shows two representative strategy predictions that illustrate both ends of the accuracy spectrum. In the VER example (Round 1, Bahrain), the predicted pit window correctly brackets the actual pit stop at lap 38 and the cumulative delta chart confirms the crossover is clearly detected. In the NOR example (Round 3, Australia), the actual pit occurred on lap 13—an early undercut

driven by track position—while the model predicted a window beginning around lap 35. This race represents the primary failure mode: reactive, non-degradation-driven pit stops that no lap time model can anticipate.



(a) VER, Round 1 (Bahrain): correct prediction. Predicted pit window (green) brackets the actual stop at lap 38.



(b) NOR, Round 3 (Australia): miss. Actual stop on lap 13 was a tactical undercut; model predicted degradation-driven window at lap 35+.

Figure 3: Representative strategy predictions. Top panel: actual vs. predicted lap times and predicted pit window. Bottom panel: cumulative time delta (positive = staying out is faster, negative = pitting is faster).

4.3 Analysis

4.3.1 GRU Dominates LSTM Variants

The GRU outperformed all LSTM variants in every experimental configuration (Table 3, Figures 1–2). The margin is large: the GRU achieved a 38–40% MAE improvement over the baseline, while the best-case LSTM only achieved 4–31%. Two factors explain this:

Parameter efficiency: GRU has 39,425 parameters vs. 52,545 for LSTM (a 25% reduction), lowering overfitting risk on a 28,000-sample training set. The consistent performance of high-dropout LSTM (+31.5%) confirms that overfitting,not capacity, is the binding constraint.

Short sequence suitability: With a 10-lap window, the cell state memory advantage of LSTM is irrelevant. Degradation within a 10-lap stint is approximately monotonic; complex long-range dependencies are absent. GRU's simpler update mechanism is well-matched to this structure.

4.3.2 *Attention Fails at Short Sequences*

LSTM with additive attention performed worst of all variants (-22.5% vs. baseline). With only 10 timesteps, the variation across hidden states is too small for the attention layer to learn a discriminative weighting; weights collapse toward uniform, and the additional non-linearity introduces noise. This is consistent with prior work showing attention benefits sequences of 20+ timesteps [5].

4.3.3 *Delta Target is the Key Representation*

Switching the training target from absolute lap time to lap-to-lap delta (Δt) roughly doubled strategy accuracy (9.5% $\rightarrow$ 19.0%) on the original 42-evaluation pre-Phase-1 subset, without changing the model architecture. The 9.5% baseline is the absolute-lap-time GRU result reported in Table 7 (Run 2 GRU); 19.0% is the same architecture retrained on the delta target before any of the strategy-level phases in Table 4 were applied. The improvement is conceptually clean since absolute lap time prediction forces the model to jointly explain baseline circuit pace and the marginal degradation signal, which are entangled in the normalized feature space. The delta formulation isolates the degradation rate as the sole target, which is what the strategy simulator actually needs.

Figure 4 shows GRU predictions against held-out 2024 lap time sequences. The time series panel (top) shows the model tracking the degradation slope within each stint, including step changes at pit stops. The scatter plot (bottom) reveals a slight underprediction bias at high normalized lap times—corresponding to heavily degraded tires late in stints—which is where the pit crossover signal is most critical.

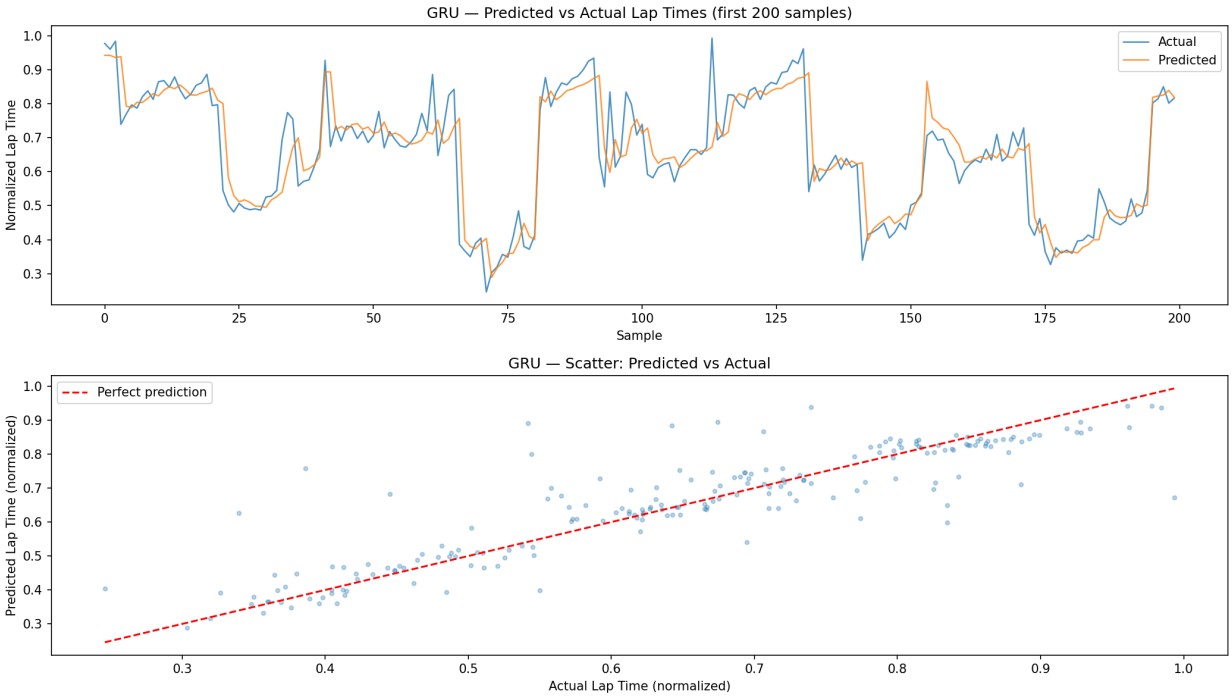


Figure 4: GRU predictions on held-out 2024 sequences (first 200 samples). Top: predicted vs. actual lap time over time, showing accurate tracking of within-stint degradation slopes. Bottom: scatter of predicted vs. actual; the slight underprediction at high lap times indicates the model conservatively underestimates late-stint degradation.

4.3.4 *MAE and Strategy Accuracy Are Decoupled*

A striking pattern across three separate experiments is that lower test MAE does not imply higher strategy accuracy (Table 7). Run 3 achieves the best MAE improvement (+40%) but produces 0% strategy accuracy. Run 6 (which adds the globally-normalized `TrackTemp_global` feature alongside the existing per-race `TrackTemp`, giving the model both a circuit-relative and an absolute temperature signal) achieves better MAE than the best strategy-level configuration but yields only 2.4% strategy accuracy.

Table 7: MAE vs. Strategy Accuracy Divergence

Configuration	Change	MAE Improvement	Strategy Accuracy
Run 2 GRU	Baseline GRU	+38.5%	9.5%
Run 3 GRU	SC filtering	+40.0%	0%
Run 6 GRU	Global TrackTemp	+41.8%	2.4%
Phase 2 compound GRU	Compound split	—	**61.9%**

This divergence occurs because MAE measures accuracy on the test lap time distribution, while strategy accuracy measures timing quality on a different task: detecting the crossover point where pitting becomes net beneficial. A model that explains circuit-level temperature signals well (Run 6) may have lower MAE, but predict degradation slopes that are biased by those irrelevant signals, leading to systematically wrong pit timing.

4.3.5 *Compound-Specific Models Are the Key Breakthrough*

The single largest accuracy improvement is achieved in Phase 2 by splitting training by tire compound ($+26.2$ percentage points over Phase 1). A shared GRU must average degradation behavior across SOFT, MEDIUM, and HARD compounds, whose degradation rates differ by a factor of 2–3. A compound-specific model sees only the degradation behavior relevant to the current tire, producing sharper predictions near the pit window crossover.

4.3.6 Negative Experiments

Two experiments were reverted after causing regression:

Phase 3 — Minimum first-pit constraint: A hard lower bound of $\max(15, 0.2 \times L_{\text{total}})$ laps was imposed on the first pit recommendation to eliminate spurious early-pit predictions. This yielded 59.5% (vs. 61.9%), as it broke two races where early pits (laps 13–15) were strategically correct (Japan undercut, China sprint). A fixed minimum is too blunt without circuit-specific context.

Phase 4 — Longer sequence (seq=15): Increasing the window from 10 to 15 laps reduced training sequences by 15% (fewer stints long enough to yield 15-lap windows) and shifted the model's first-prediction opportunity later in each stint, producing an accuracy of 50.0%.

4.3.7 Per-Driver Spread Reveals Identity Bottleneck

The 30 percent spread in Table 6 (HAM 40% $\rightarrow$ RUS 70%) cannot be explained by team effects—same-team pairs diverge significantly (HAM 40% vs. RUS 70% on Mercedes; LEC 45% vs. SAI 60% on Ferrari). This points to driver-specific stint patterns that the current model cannot capture. Russell ran notably more conservative stint strategies in 2024 (later first pits, consistent MEDIUM stints) that align well with the model's predictions; Hamilton's more aggressive tire management and frequent off-sequence stops are harder to predict.

The NOR miss in Figure 3b is emblematic of this class of failure. The model correctly identifies when degradation would warrant a stop, but cannot anticipate that the team will sacrifice lap time to execute a tactical undercut instead. The model has no driver or team identity feature. Adding driver one-hot encoding or team-level embeddings would be the most impactful next step.

5. Conclusions

5.1 Summary

This project demonstrates that publicly available F1 lap time data, combined with compound-specific GRU sequence models and a simulation-based strategy evaluator, can recommend pit stop

windows with 52.5% accuracy across 122 race-driver evaluations from the 2024 season. The result represents a +21.5% improvement over the single-stop, shared-GRU baseline (Table 4). The GRU architecture with delta target, compound-specific training, and per-race normalization is the configuration that best captures the tire degradation signal relevant to strategy timing.

The 61.9% result on the original 2-driver subset (Verstappen and Norris, 42 evaluations) is the strongest single-configuration result but is best understood as an upper bound on favorable conditions; 52.5% on the expanded 122-evaluation set is the honest large-sample estimate. At the per-driver level the upper end is higher still (Russell at 70.0%, Sainz at 60.0%), but the 30% spread across drivers shows that any single accuracy figure depends heavily on which drivers are evaluated.

Ulltimately, this approach was proved to be successful. My initial success criteria was atleast a 10% improvement in MAE over the moving average baseline. Every GRU configuration that was evluated exceeds this target by roughly 4 times.

5.2 Contributions

This project makes the following contributions:

- A full open-source pipeline for F1 pit stop strategy evaluation using only publicly available data (FastF1 API, 2022–2024 seasons).
- An empirical demonstration that compound-specific sequence modeling dramatically outperforms shared-compound models for strategy timing ($+26.2$ pp).
- A systematic ablation showing that MAE on lap time prediction is a poor proxy for strategy accuracy, with three experiments showing conflicting directions.
- Quantification of the driver identity gap: per-driver accuracy ranges from 40% to 70% in the same evaluation, setting a clear target for future work.
- Documentation of five strategy-level experimental phases including two fully reverted experiments with root-cause analysis.

5.3 Future Work

The most impactful near-term improvement is adding driver and team identity features. Even a simple driver one-hot encoding over the 20 F1 drivers would allow the model to learn driver-specific degradation tendencies and strategy preferences. Based on the Phase 5 analysis, this feature alone could close a substantial portion of the per-driver accuracy gap.

Additional directions include:

- **More training data**: Adding 2021 and 2020 season data would triple the training set. The LSTM deep ablation showed that deeper models directly benefit from more data.
- **Track identity features**: Circuit-specific embeddings would allow the model to learn that Monaco and Singapore have different degradation profiles than Bahrain and Spa.
- **Competitor position**: Real teams pit partly based on gap to the driver ahead or behind. Including relative position would move the system closer to true strategy modeling.
- **Transformer architecture**: With driver/track identity tokens, a transformer encoder [9] over the 10-lap sequence may outperform the GRU, as cross-stint attention becomes meaningful once distinct sequence types are identifiable.

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A. Experiment Reproducibility

All code is organized as follows:

- `data/pipeline.py`: FastF1 data fetching and feature engineering
- `data/preprocessing.py`: Sequence building, normalization, compound splitting
- `train.py`: Model training for all configurations
- `strategy/pit_optimizer.py`: Strategy simulation and evaluation
- `models/`: LSTM, GRU, and attention model definitions
- `results/`: Saved model weights and evaluation CSVs

To reproduce the Phase 5 results from scratch:

1. `python data/preprocessing.py` — regenerates training and test tensors

2. `python train.py` — trains the three compound-specific GRU models
3. `PYTHONPATH=. python strategy/pit_optimizer.py` — runs strategy evaluation on all 6 drivers across the 2024 season

Training requires approximately 10 minutes on Apple M-series hardware (MPS backend). The strategy evaluation requires approximately 40–60 minutes due to FastF1 data loading per race.